

# Investigating the Relationship Between Cranial Capacity and Evolutionary, Biological, and Environmental Factors in Hominins: A Bayesian Approach

## 1.ABSTRACT

Cranial capacity, often used as a proxy for brain size, plays a critical role in understanding the cognitive and evolutionary development of hominins. As hominins evolved, increases in brain size were accompanied by behavioural shifts, technological advancements, and changes in diet. Understanding the relationships between cranial capacity and key factors such as height, evolutionary time, technology, and diet can provide deeper insights into the evolutionary pressures shaping cognitive development. This study employs a Bayesian regression model to explore how these variables contribute to cranial capacity. The Bayesian approach allows for the incorporation of uncertainty in parameter estimation, offering a more robust analysis than traditional methods. Key findings reveal significant positive associations between cranial capacity and both height and technology use, while time exhibits a negative association, reflecting the gradual increase in brain size across evolutionary periods. Interestingly, while an omnivorous diet showed a negative association with brain size, diets rich in hard fruits

displayed positive correlations. These findings highlight the complex interplay between body size, technological innovations, and dietary adaptations in hominin evolution. This study provides valuable insights for anthropology and evolutionary biology, offering a refined statistical approach to analysing evolutionary data.

**KEYWORDS:** Cranial Capacity Evolution, Bayesian Statistical Analysis, Hominin Cognitive Development, Environmental Factors in Evolution, MCMC Sampling, JAGS Analysis, Diet Cognition Relationship.

## 2.INTRODUCTION

Cranial capacity, often used as a proxy for brain size, is one of the most critical indicators of cognitive and evolutionary development in hominins. As hominins evolved, significant changes in behaviour, technological innovation, and dietary patterns coincided with increases in brain size, suggesting a link between cranial capacity and these evolutionary factors. Understanding the factors that contributed to brain expansion is crucial for interpreting the evolutionary trajectory of hominins. However, the nature and strength of the

relationships between cranial capacity and factors like height, diet, and technology remain underexplored, with much debate surrounding whether these factors were drivers of brain expansion or consequences of cognitive evolution (Ash & Gallup, 2007).

Previous studies have primarily relied on traditional regression models, examining correlations between cranial capacity and various predictors. For instance, high-protein diets and technological advancements have been suggested as potential contributors to brain growth (Beaudet et al., 2019). However, these methods often fail to account for the uncertainty inherent in estimating these relationships, potentially limiting our understanding of their true impacts. This study aims to address this gap by applying a Bayesian regression model, which offers a more nuanced and probabilistic approach to understanding how factors like height, technology, diet, and time influence cranial capacity.

The objective of this study is to quantify the relationships between cranial capacity and these evolutionary factors, while accounting for uncertainties often neglected in traditional analyses (Jreich et al., 2022). By utilizing a Bayesian approach, this study provides a refined framework for analysing cranial capacity trends across different hominin species, contributing to the broader understanding of human evolution.

This report is structured to provide a comprehensive analysis of the

relationships between cranial capacity and key evolutionary factors. The Literature Review covers existing research on cranial capacity, identifying gaps in current methodologies. The Research Questions section formulates hypotheses regarding the associations between cranial capacity and variables such as height, time, technology, and diet. The Research Design/Methods section explains the use of Bayesian regression and the rationale for employing MCMC sampling to account for uncertainty. The Results section presents posterior distributions, credible intervals, and effective sample sizes for the regression parameters. The Discussion evaluates the implications of the findings, addresses limitations, and suggests avenues for future research. Finally, the Conclusion summarizes the key findings and their relevance to understanding hominin evolution. This structure ensures a thorough exploration of how biological and environmental factors are connected to cranial capacity while addressing gaps in the literature.

### **3.LITERATURE REVIEW**

#### **3.1 Cranial Capacity and Evolutionary Trends**

Research on the evolution of cranial capacity in hominins has provided significant insights into the adaptive pressures influencing brain size. Ash and Gallup explored the impact of paleoclimatic variation on brain expansion, showing that temperature changes played a critical role in driving

the evolutionary increase in cranial capacity. Their findings suggest that nearly 52% of the variation in brain size could be attributed to temperature fluctuations, with equatorial distance also contributing to cranial size differences(Ash & Gallup, 2007).

Bailey and Geary expanded on this by examining the role of social competition alongside climatic pressures. Their multivariate analysis indicated that population density and temperature variation were independent contributors to the evolution of brain size, with social factors likely playing a more dominant role in driving brain expansion in early hominins(Bailey & Geary, 2009) .

### **3.2 Technological Advancements and Brain Size**

Tool use and dietary shifts have been central to the understanding of brain evolution. Braun et al found that early hominins incorporated aquatic animals rich in brain-growth nutrients into their diet, suggesting that access to such resources was a key adaptation that allowed for larger brain sizes. This evidence aligns with the broader view that technological and dietary shifts co-evolved with cognitive abilities, providing the necessary energy to sustain increased brain size(Braun et al., 2010).

Beaudet et al reviewed the evolution of the modern human brain, emphasizing that brain size and shape changes in the hominin lineage were driven by both technological advancements and complex social behaviours. Their review

highlighted the challenges of addressing these evolutionary changes using frequentist methods, calling for more probabilistic approaches to account for uncertainty(Beaudet et al., 2019).

### **3.4 Social and Environmental Pressures**

Social networks and cooperation may have further contributed to the development of larger brains. Apicella et al compared the social networks of Hadza(A group of tribe in African continent) hunter-gatherers to modern social structures, concluding that cooperative behaviour may have emerged as an adaptive response to environmental and social challenges, thus driving brain expansion in early human populations(Apicella et al., 2012).

Neubauer et al took a different approach by examining the evolution of brain shape, showing that modern human brain globularity evolved gradually, paralleling behavioural modernity. Their findings suggest that brain reorganization, in addition to size increase, played a significant role in hominin evolution(Neubauer et al., 2018).

### **3.5 Methodological Advances in Brain Evolution Research**

Traditional methods for studying brain size evolution have relied heavily on regression models, which often fail to capture the inherent uncertainties in evolutionary data. De Miguel and Henneberg highlighted the challenges of

accurately estimating cranial capacity from fossil records, noting the large error margins associated with reconstruction methods. They argued that much of the variability in cranial capacity estimates stems from the limitations of the techniques used (De Miguel & Henneberg, 2001).

Recent advances in Bayesian methods have provided a framework for addressing these uncertainties. reviewed Bayesian selection methods, emphasizing their utility in evolutionary studies. They found that Bayesian models could better account for the variability in data, particularly in cases involving categorical predictors and uncertain phylogenetic relationships.

### **3.6 Gaps in the Literature**

While significant progress has been made in understanding the factors influencing cranial capacity, several knowledge gaps remain. First, there is still uncertainty surrounding the directional relationships between environmental pressures, social complexity, and brain size. Bednarik proposed that brain atrophy, observed in the final Pleistocene and Holocene periods, may have been driven by the use of exograms external memory aids suggesting that cultural evolution played a role in recent reductions in brain size. This hypothesis challenges the long-standing view that brain expansion was a continuous process throughout hominin evolution (Bednarik, 2014).

Moreover, while Williams and Hill highlighted the importance of diet, particularly meat and vitamin B3 intake, in brain development, the exact mechanisms linking dietary changes to cognitive evolution remain unclear. The role of climate, as proposed by (Ash & Gallup, 2007), requires further investigation, especially in relation to shorter-term climate fluctuations and their effects on brain size (Williams & Hill, 2017).

## **4. RESEARCH QUESTIONS**

This study seeks to explore the relationships between cranial capacity and a range of evolutionary, biological, and environmental factors in hominins, focusing on the interplay between diet, technology, height, and time. The central questions guiding this research are:

- 1. What is the nature of the relationship between cranial capacity and dietary patterns in hominins?**

This question addresses whether a high-protein diet contributed to brain expansion or whether a larger brain facilitated more advanced dietary behaviours, including hunting and meat consumption.

- 2. How do technological advancements (e.g., tool use) correlate with increases in cranial capacity?**

This question explores the direction of causality: whether the development of tools and

technology spurred brain growth, or if an expanding brain allowed for more complex technological innovations.

3. **What role do evolutionary time periods and environmental adaptations play in cranial capacity development?**

This question examines whether hominins in different environmental conditions, such as colder climates, exhibit variations in cranial capacity due to adaptive pressures.

4. **To what extent does height (as a proxy for body size) contribute to variations in cranial capacity?**

This question investigates the relationship between cranial capacity and body size, seeking to clarify whether larger bodies naturally evolved with larger brains or if other factors are more significant.

**5. Scope of the Research:**

This study focuses on the relationships between cranial capacity and four key predictors: diet, technology (Tecno), height, and time in hominins. It uses a Bayesian regression framework to account for uncertainty in estimating these relationships and to explore the probabilistic relationship between the variables.

- **Diet:** The study includes omnivorous, carnivorous, and other diets to understand how

nutrition influenced brain development. It is a categorical variable which contains five different categories, which are “Carnivorous”, “Omnivorous”, “Hard Fruits”, “Soft Fruits” and “Dry Fruits”.

- **Technology:** Technological advancements, measured by the presence of tools, are analysed to determine how cognitive development might have spurred or resulted from tool use. It is a categorical variable which has three categories, which are “No”, “Yes” and “Likely”.
- **Height:** Height is used as a proxy for body size, enabling the analysis of how physical evolution interacted with brain size. It is a continuous variable which denotes the average height or stature of each hominid species which is measured in cm.
- **Time:** Time periods in hominin evolution are incorporated to account for the temporal aspect of brain growth and its relationship with evolutionary pressures. It is a continuous variable which indicates the time period during which each hominid species lived, it is measured in mya (Millions of years ago).

This study employs a Bayesian regression model, offering a probabilistic framework that

incorporates prior knowledge about hominin evolutionary history and computes posterior distributions for the coefficients associated with each predictor. The analysis uses quantitative data on cranial capacity, diet, technology, height, and time, providing a robust statistical understanding of these relationships while quantifying the uncertainty that traditional models often struggle to address. While linear regression can provide useful insights, it assumes fixed parameter estimates and lacks the ability to incorporate uncertainty directly into the model. In contrast, Bayesian approaches account for uncertainty by generating posterior distributions for model parameters. The study limits its focus to cranial capacity as a proxy for brain size, excluding other cognitive markers such as behavioural complexity, language, or social structures.

## **6. RESEARCH DESIGN**

The research was structured to answer key questions surrounding the relationships between cranial capacity and evolutionary factors, including diet, technology, height, and time in hominins. The research followed a sequential plan, starting with data acquisition and progressing through data preprocessing, model refinement, and final analysis using Bayesian regression. The main activities and steps involved were:

### **6.1 Data Collection and Preprocessing:**

For this study, the dataset was sourced

from Kaggle and underwent a series of preprocessing steps to ensure its suitability for Bayesian regression analysis. First, categorical variables such as diet and technology were converted into factors and dummy variables for inclusion in the regression model. The date variable, representing evolutionary time periods, was transformed into negative values to represent the number of years ago, allowing it to be used as a continuous variable. The distribution of numerical variables was checked to ensure normality, and correlations between key variables were analysed to identify any significant relationships prior to modelling. To standardize the scale of numerical variables, such as height, time and cranial capacity, were standardized to have a mean of zero and a standard deviation of one, improving the model's performance. Data was checked for null values to ensure a complete dataset. These preprocessing steps were critical in preparing the dataset for the Bayesian regression analysis, facilitating a robust investigation into the relationships between cranial capacity and evolutionary factors like diet, technology, height, and time.

### **6.2 DATA LINK: [KAGGLE](#)**

#### 6.4 Model Selection: Bayesian Regression

Bayesian regression was selected as the most appropriate method for this study because of its ability to allow for the inclusion of prior distributions based on existing research, making it easier to account for the uncertainties inherent in fossil data. The posterior distributions generated in Bayesian analysis offer a probabilistic understanding of the relationships between variables, which is especially valuable in evolutionary studies (Jreich et al., 2022). Bayesian regression can accommodate models with multiple interactions, such as the interplay between diet and technology, while still providing reliable estimates.

In the JAGS implementation, cranial capacity was modelled as the dependent variable, and diet, technology, height, and time were the independent variables. Priors for the model coefficients were based on previous studies, with vague priors used for parameters where no strong prior knowledge existed. MCMC sampling was applied, with each chain running for a sufficient number of iterations to ensure convergence and generate reliable posterior estimates.

#### 7.MODEL:

$$\text{Cranial Capacity}_i \sim N(\mu_i, \tau)$$

The mean cranial capacity ( $\mu_i$ ) is modelled as

$$\begin{aligned} \mu_i = & \beta_0 + \beta_1 * \text{Time}_i + \beta_2 * \\ & \text{Height}_i + \beta_3 * \text{Tecno}_2 + \beta_4 * \\ & \text{Tecno}_3 + \beta_5 * \text{Diet}_2 + \beta_6 * \text{Diet}_3 + \\ & \beta_7 * \text{Diet}_4 + \beta_8 * \text{Diet}_5 + \varepsilon_i \end{aligned}$$

Where:

- Tecno2 = “Yes”
- Tecno3 = “Likely”
- Diet2 = “Soft fruits”
- Diet3 = “Omnivore”
- Diet4 = “Carnivore”
- Diet5 = “Hard Fruits”

##### 7.1 Strengths of the Method:

Bayesian regression is flexible enough to model complex relationships and handle uncertainties in data, making it ideal for evolutionary studies where fossil records are incomplete. The posterior distributions provide insights into the probability of relationships, which is more informative than the point estimates produced by traditional regression methods.

##### 7.2 Weaknesses of the Method:

Bayesian methods, especially MCMC sampling, are computationally demanding. Achieving convergence for complex models may require extensive iterations and computational resources. The results can be sensitive to the choice of prior distributions, which can lead to biased estimates if not properly chosen. This was mitigated in the study by conducting sensitivity analyses

during the pilot phase, but remains a potential limitation.

### 7.3 Alternative Methods Considered

- **Traditional Linear Regression:** Initially, linear regression was considered for its simplicity and ease of interpretation. However, it was rejected due to its inability to account for the uncertainty inherent in evolutionary data and its limitations in modelling complex interactions between variables.
- **Hierarchical Models:** Hierarchical Bayesian models were considered, given their ability to model evolutionary relationships across species. However, the available data did not include sufficient phylogenetic information to justify the use of a hierarchical model. Additionally, the computational complexity and time required to fit such models outweighed their potential benefits for this study.

### 7.4 Ethical Considerations

The dataset used for this research involved fossil measurements and did not involve living subjects, meaning that traditional ethical concerns such as informed consent or privacy were not applicable. However, the ethical handling of data sources and accurate reporting were critical aspects of the study. All data were sourced from publicly available databases and

referenced appropriately, ensuring compliance with academic standards of integrity.

### 7.5 Methodological Challenges and Refinement

Several challenges arose during the implementation of the Bayesian model, particularly in selecting appropriate prior distributions and achieving model convergence. Early tests revealed that vague priors contributed to slow convergence within the MCMC chains. The priors were refined by incorporating insights from relevant literature and domain knowledge. Instead of using non-informative or vague priors, weakly informative priors were selected to provide mild constraints on the parameters, guiding the model towards more reasonable estimates without introducing bias. This approach improved the stability and efficiency of the Markov Chain Monte Carlo (MCMC) sampling process, ensuring faster convergence and more reliable posterior distributions (van de Schoot et al., 2021).

Additionally, convergence diagnostics such as trace plots and the Gelman-Rubin statistic were used throughout the model refinement process to assess whether the MCMC chains were mixing well. The model ran 10,000 iterations per chain across three parallel chains, ensuring that the sampling captured the posterior distributions effectively. A burn-in period of 2,000 iterations was employed to allow the chains to stabilize by discarding the initial



samples influenced by the starting values. After burn-in, a thinning factor of 10 was applied, meaning only every 10th sample was retained to minimize autocorrelation between consecutive samples and improve the quality of the posterior estimates. The total number of effective samples across the three chains provided reliable estimates for the posterior distributions. These steps, including careful diagnostics, ensured that the final estimates were based on well-mixed chains, improving both the reliability and robustness of the results

## 8. Application of the Research Method

The research method applied in this study focused on using a Bayesian regression model to investigate the relationships between cranial capacity and key factors such as height, time, diet, and technology in hominins. The model was implemented using JAGS, with MCMC sampling used to generate posterior distributions for each parameter after an initial burn-in period. The model was tested using a train-test split strategy, where 70% of the data was used for training and 30% for testing, providing robust evidence of the model's performance.

### 8.1 Model Testing and Performance

The model performed strongly on the test set, demonstrating its reliability in predicting cranial capacity. The key performance metrics include:

- **R-squared value:** The model achieved an R-squared of 0.84, explaining 84% of the variance in

cranial capacity, indicating a strong fit between the model and the observed data.

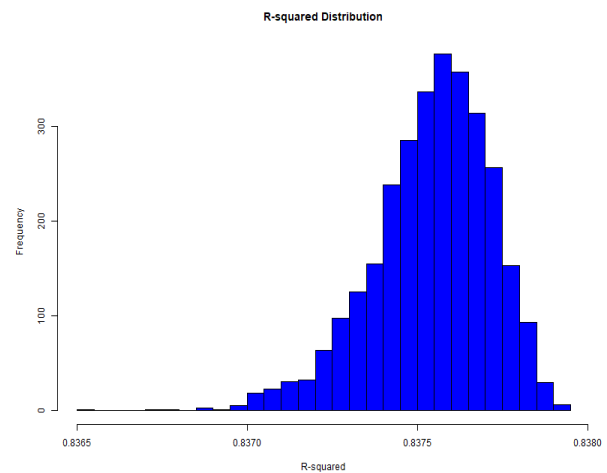


Figure 1: R-squared Distribution

- **Mean Absolute Error (MAE):** The MAE was 0.32, indicating that, on average, the predicted values were close to the actual values.
- **Mean Squared Error (MSE):** The MSE was 0.16, demonstrating that the model had a low level of average squared error, further confirming its predictive power.
- **Residual analysis:**

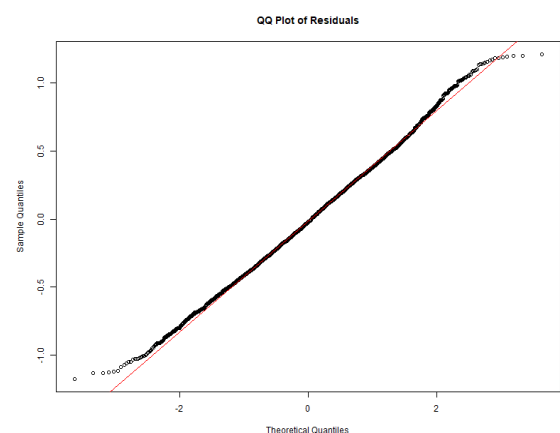


Figure 2: Normality Plot

Figure 2 showed no significant patterns, indicating that the residuals were randomly distributed and that the model was well-fitted without systematic errors.

## 8.2 Bayesian Regression Results

| PREDICTOR                    | 95% CREDIBLE INTERVAL (LOWER) | 95% CREDIBLE INTERVAL (UPPER) |
|------------------------------|-------------------------------|-------------------------------|
| Time (Beta 1)                | -0.157                        | -0.1278                       |
| Height (Beta 2)              | 0.491                         | 0.516                         |
| Technology:"YES" (Beta 3)    | 1.032                         | 1.107                         |
| Technology:"Likely" (Beta 4) | 0.381                         | 0.484                         |
| Soft Fruits Diet (Beta 5)    | 0.053                         | 0.114                         |
| Omnivorous Diet (beta6)      | -0.577                        | -0.477                        |
| Carnivorous Diet (beta7)     | -0.025                        | 0.097                         |
| Hard Fruits Diet (beta8)     | 0.365                         | 0.423                         |

Table 1: CREDIBLE INTERVALS OF THE PREDICTORS

The Bayesian regression model provided probabilistic insights into the relationships between cranial capacity and evolutionary factors, expressed through posterior distributions and credible intervals. Height showed a significant positive association with cranial capacity, with a posterior mean of 0.503 and a 95% credible interval of [0.491, 0.516]. This suggests there is a 95% probability that the true effect of height on cranial capacity lies within this interval, indicating taller hominins were likely to have larger brains, with a

high level of certainty in this association.

Technological advancements, particularly tool use, also exhibited a positive association, with a posterior mean of 1.070 and a 95% credible interval of [1.032, 1.107]. This implies a 95% probability that the true impact of tool use on brain size falls within this range, supporting the hypothesis that technology contributed meaningfully to brain growth. Furthermore, technology categorized as "likely" (beta4) had a positive effect with a mean of 0.428 and a 95% credible interval of [0.381, 0.484], reinforcing the significance of technological advancements in brain evolution.

The time factor had a posterior mean of -0.142 with a 95% credible interval of [-0.157, -0.127], indicating a 95% probability that cranial capacity increased over evolutionary periods, with a negative slope showing that more recent hominins had larger brains. Dietary patterns presented more complex findings. An omnivorous diet was associated with a negative effect on cranial capacity (mean: -0.523, 95% credible interval: [-0.577, -0.477]), suggesting a 95% probability that the omnivorous diet had a negative influence on brain size, challenging traditional assumptions about the role of diverse diets in cognitive development. In contrast, a soft fruit diet showed a positive association with cranial capacity (mean: 0.083, 95% credible interval: [0.053, 0.114]), with a 95% probability that the effect is

The error term ( $\sigma$ ) had a mean of 0.401 with a 95% credible interval of [0.395, 0.406], representing the standard deviation of the residuals. This indicates the average deviation of observed cranial capacity values from the predicted values by the model. A smaller value of “ $\sigma$ ” suggests that the model's predictions are closer to the observed values. The 95% credible interval indicates that, with 95% certainty, the true standard deviation of the residuals lies between 0.395 and 0.406, capturing the uncertainty associated with the unexplained variability in the model.

One of the most unexpected findings was the negative association between an omnivorous diet and cranial capacity. While it has been traditionally assumed that a diet rich in protein and diverse food sources would drive brain expansion, the results challenge this notion and suggest that other factors, such as technology and height, played a

## 6.3 Visualization



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display much larger brain sizes. The color-coded points show that species with larger cranial capacities evolved closer to the present, suggesting a general increase in brain size over time as hominins adapted and developed more complex behaviours and cognitive abilities.

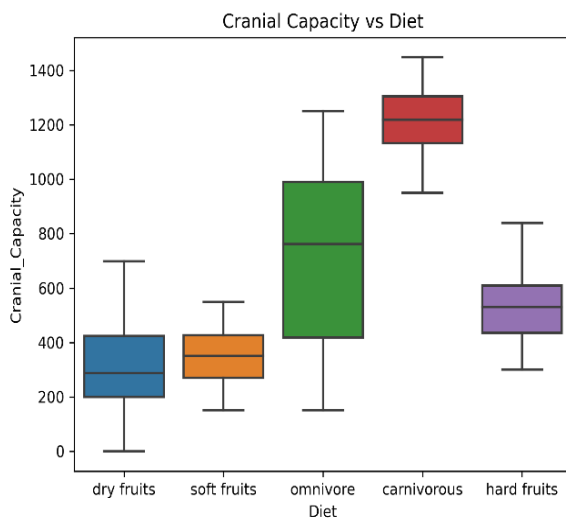


Figure 4: Cranial Capacity and Diet Types

Figure 4 illustrates the relationship between cranial capacity and diet types across various hominin species. The horizontal axis represents different dietary categories: dry fruits, soft fruits, omnivore, carnivorous, and hard fruits, while the Y-axis indicates cranial capacity in cubic centimeters. The median cranial capacities for each diet type vary significantly. Hominins with an omnivorous diet exhibit the largest cranial capacities, with a broad range, followed closely by those with a carnivorous diet. In contrast, diets based on dry fruits, soft fruits, and hard fruits are associated with smaller cranial capacities, with lower variability in cranial capacity among these groups.

This suggests that a more varied or protein-rich diet, particularly omnivorous and carnivorous, may have been associated with larger brain sizes, while fruit-based diets correspond to smaller cranial capacities.

### 9.1 Analysis Linked to Research Problem

The model successfully addressed the research question by revealing significant associations between cranial capacity and evolutionary factors such as height, technology, and time. The results supported the idea that technological advancements and body size were key drivers of cranial expansion, while the role of diet was more complex and less impactful than expected. The Bayesian approach allowed for a more nuanced understanding of these relationships by accounting for uncertainty and offering credible intervals for all estimates.

## 10. CONCLUSION

This study explored the intricate relationships between cranial capacity and key evolutionary factors: height, technological advancements, time, and diet, using a Bayesian regression framework. The results demonstrated that height and technological use had strong positive associations with cranial capacity, indicating that body size and cognitive complexity, reflected through tool use, were significant contributors to brain expansion in hominins. The findings further revealed that cranial capacity increased over evolutionary time, consistent with the idea of gradual

cognitive development in early humans. Contrary to traditional beliefs, the study found a negative association between omnivorous diets and cranial capacity, suggesting that diet may have played a more complex and less dominant role than previously thought. This research contributes to the broader understanding of hominin evolution by clarifying the probabilistic relationships between brain size and its evolutionary predictors.

### **11.Limitations**

Despite its contributions, the study has several limitations. One challenge was the computational demands of the Bayesian regression model, particularly with MCMC sampling, which required significant time and resources for convergence, potentially limiting the scalability of this approach for larger datasets. Additionally, the model exhibited sensitivity to priors, meaning that while vague priors were used to avoid bias, the selection of different priors could influence the outcomes, requiring careful consideration in future analyses. Another limitation is the correlational nature of the study, as it identified significant relationships between cranial capacity and the predictors but could not establish causality. The associations observed, particularly between technology, diet, and cranial capacity, remain correlational, highlighting the need for further investigation to determine causal directions. Finally, the study focused on four main predictors height, time, technology, and diet excluding

other relevant variables such as social structures, climate, and genetic diversity, which may have also influenced cranial evolution. These limitations underscore the need for future research to address these gaps and improve the robustness of the findings.

### **12.Future Work**

Future research can expand on this study by exploring cranial capacity trends in closely related species, such as non-hominin primates (e.g., chimpanzees, gorillas) and extinct human relatives like Neanderthals and *Homo erectus*, to identify evolutionary pressures unique to hominins. Comparative studies with large-brained mammals, such as dolphins and elephants, or cognitively advanced birds like crows, could reveal parallel evolutionary patterns, offering insights into the role of ecological niches in brain development.

Additionally, incorporating environmental data like climate and social variables like group size will provide a more holistic view of cognitive evolution. Advanced methodologies, such as hierarchical Bayesian models and causal inference techniques, can help uncover the directionality of relationships between brain size and predictors like diet and technology. Furthermore, efforts to improve computational efficiency through methods like Variational Inference, along with the use of 3D modelling for fossil measurements, will enhance data accuracy and scalability. Interdisciplinary collaboration and the

development of open-access databases will foster transparency and encourage further research, ensuring continued progress in understanding the complexities of brain evolution across species.

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